

Satyam Awasthi

Researcher & Software Engineer

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Research Summary

Researcher working at the intersection of **neural interfaces**, **physiological sensing**, **learning**, and **control**. Current work spans **EEG-based brain-computer interfaces** and **robust output-feedback model predictive control**.

Research Interests

Neural Interfaces, Brain-Computer Interfaces, Physiological & Wearable Sensing, Human-Machine Interaction, Machine Learning for Biosignals, Robust State Estimation, and Model Predictive Control.

Research Experience

Autonomous Systems Group, Imperial College London

Research Collaboration

Jul 2026 – Present

Advisor: [Dr. Johannes Köhler](#)

- Investigating **state-dependent estimation and tracking-error bounds** for nonlinear robust output-feedback MPC with state-dependent process and measurement uncertainty.
- Developing coupled observer and tracking-error tube propagation using **contraction-based certificates** for robust constraint tightening and uncertainty-aware trajectory planning.

Four Eyes Laboratory, University of California, Santa Barbara

Graduate Researcher

Sep 2021 – Jun 2023

Advisor: [Prof. Tobias Höllerer](#); Co-Advisor: [Prof. Michael Beyeler](#)

- Developed **EyeTTS**, an open-source framework for evaluating and calibrating eye tracking during mixed-reality locomotion, resulting in first-author publications at IEEE ISMAR Adjunct and IEEE VR Workshops.
- Investigated how **oculomotor behaviors—including fixation, saccades, and smooth pursuit**—affected gaze-tracking precision and temporal response, including interactions with device-side signal filtering during locomotion.
- Designed human-subject experiments and analyzed gaze-tracking accuracy, calibration drift, and motion-dependent sensing error across multiple mixed-reality devices.
- Released a reproducible **calibration framework** and **multi-device eye-tracking dataset** containing gaze trajectories, calibration measurements, and locomotion data.

Autonomous Ground Vehicle & Swarm Robotics Group, IIT Kharagpur

Undergraduate Researcher

Jan 2016 – Dec 2018

- Contributed perception, multimodal state-estimation, and navigation components to **Eklavya 6.0**, IIT Kharagpur's autonomous ground vehicle that finished **Runner-Up at IGVC 2018**.

Publications

Peer-Reviewed Conference Papers

SmartWatch for Wall Writing: Real-time Transcription of Wall Writing from Inertial Sensing

Snigdha Das, **Satyam Awasthi**, Abdul Shamnar P, Pradipta De, Sandip Chakraborty, and Bivas Mitra.

COMSNETS, 2022.

DOI: [10.1109/COMSNETS53615.2022.9668531](https://doi.org/10.1109/COMSNETS53615.2022.9668531)

Workshop & Poster Publications

Eye Tracking Performance in Mobile Mixed Reality

Satyam Awasthi, Vivian Ross, Sydney Lim, Michael Beyeler, and Tobias Höllerer.

IEEE VR Workshops (VRW), 2024.

DOI: [10.1109/VRW62533.2024.00321](https://doi.org/10.1109/VRW62533.2024.00321)

EyeTTS: Evaluating and Calibrating Eye Tracking for Mixed-Reality Locomotion

Satyam Awasthi, Vivian Ross, Michael Beyeler, and Tobias Höllerer.

IEEE ISMAR Adjunct, 2023.

DOI: [10.1109/ISMAR-Adjunct60411.2023.00104](https://doi.org/10.1109/ISMAR-Adjunct60411.2023.00104)

Selected Research Projects

EEG-Based BCI for 3D Navigation

2026 – Present

Independent Brain-Computer Interface Project

- Developing a **4-channel EEG acquisition and decoding pipeline** for real-time control of an interactive 3D navigation environment.
- Building an initial **SSVEP-based intent-decoding system** using neural-signal preprocessing and spectral / correlation-based features for discrete movement commands.
- Investigating confidence thresholds and temporal evidence accumulation to characterize the trade-off between **decoding reliability and command latency**.

SmartWatch for Wall Writing

2019 – 2022

Wearable Inertial Sensing for Human-Computer Interaction

- Contributed to a wearable sensing system for **real-time transcription of free-form wall writing** from smartwatch inertial measurements.
- Explored the use of human wrist-motion signals as an unobtrusive input modality for recognizing physical writing gestures.

JitterNot

Jan 2022 – Mar 2022

Learning-Based Model Predictive Control for Adaptive Video Streaming (UCSB)

- Developed an LSTM-based predictor for network throughput to support closed-loop decisions under non-stationary network dynamics.
- Designed a receding-horizon **model predictive control (MPC)** framework for adaptive bitrate selection under bandwidth constraints.

In Motion — Wearable Human Activity Recognition

Nov 2019

DeepConvLSTM Modeling of Inertial Sensor Signals (IIT Kharagpur)

- Designed a streaming inference pipeline for **high-frequency IMU signals** enabling temporal human-activity classification.
- Implemented a DeepConvLSTM architecture combining convolutional feature extraction with recurrent temporal modeling of wearable sensor sequences.

Education

University of California, Santa Barbara (UCSB)

Santa Barbara, CA

M.S. in Computer Science (GPA: 3.91/4.00)

Sep 2021 – Jun 2023

Indian Institute of Technology, Kharagpur (IIT Kharagpur)

Kharagpur, India

B.Tech. in Electrical Engineering, Minor in Computer Science (GPA: 8.98/10.0)

Jul 2016 – Jul 2020

Industry Experience

Intuit Inc

Mountain View, CA · Mar 2024 – Jan 2026

Software Engineer 2

- Built **dual-stage agent pipelines** for intent routing, structured tool execution, and generative user-interface composition.
- Developed production platform infrastructure supporting large-scale client applications and internal developer workflows.

Yahoo Inc

Mountain View, CA · Apr 2023 – Nov 2023

Software Dev Engineer II (IC3)

- Developed production Android systems for Yahoo Mail, including data-driven inbox commerce and expense-management features.

Coupang Global LLC

Mountain View, CA · Jun 2022 – Sep 2022

Software Engineering Intern

- Developed user-facing search and product-discovery features for Coupang's Android commerce platform.

Samsung Research Institute

Noida, India · Sep 2020 – Sep 2021

Software Engineer (CL 2)

- Developed Android platform and device applications involving media processing, privacy, and isolated user-storage architectures.

Teaching

University of California, Santa Barbara

Graduate Teaching Assistant

Santa Barbara, CA

Sep 2021 – Jun 2023

- Led instructional labs, developed programming assignments, and supported senior capstone teams across CS189 (Capstone), CS184 (Android), CS24 (C++ Data Structures), and CS16 (Problem Solving).

Technical Skills

Programming: Python, C++, MATLAB, Java, Kotlin

Machine Learning & Signal Processing: PyTorch, NumPy/SciPy, OpenCV, Time-Series Modeling, Spectral Analysis, Digital Filtering, Wearable/Inertial Signal Processing

Control & Estimation: Model Predictive Control (MPC), Robust MPC, Contraction-Based Methods, State Estimation (EKF), Sensor Fusion, Motion Planning

Robotics & Interactive Systems: ROS/ROS2, Gazebo, RViz, Unity, OpenGL

Developer Tools: Git

Academic Service & Peer Review

Peer Reviewer, UIST 2024

ACM Symposium on User Interface Software and Technology

- Reviewed submissions on interactive systems, spatial tracking, and interface hardware.